VideoAgentTrek: Computer Use Pretraining from Unlabeled Videos

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Paper version: arXiv 2510.19488v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *raw videos contain implicit demonstrations but lack explicit action labels*

Screen tutorials show clicks and typing, but not **structured actions**. The paper learns to label them.

Motivation: three gaps

1. **Costly labels.** Manual GUI trajectories do not scale.
2. **Untapped video.** Tutorials show clicks and typed text.
3. **No action labels.** Cursor and text are visible, not tracked.

Evaluation setup

OSWorld-Verified online · AgentNetBench offline · Qwen2.5-VL-7B base · 39,000 YouTube videos · 20 / 50-step budgets · 3 runs at 20 steps · 1 run at 50 steps

- **Online:** OSWorld-Verified, 369 Ubuntu tasks. **Offline:** AgentNetBench, 100 tasks.
- **Held-out apps:** not stated in the paper.

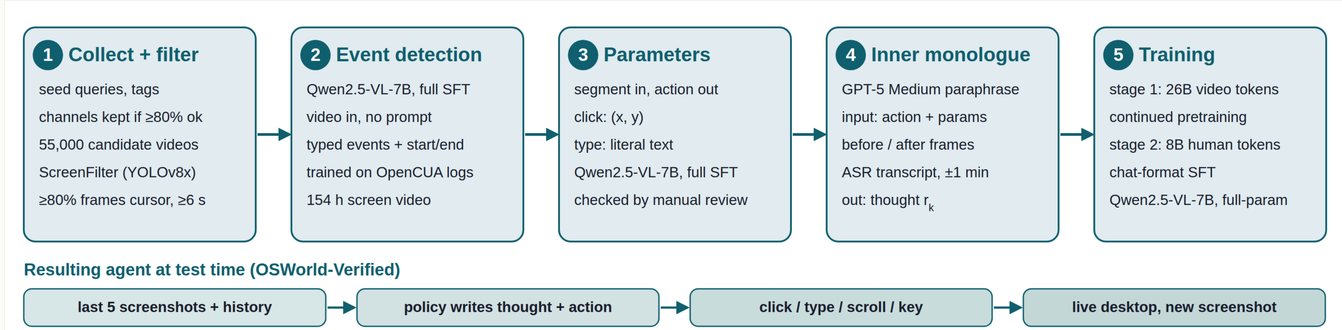
Our read (CUA Radar)

68	★★	online
score of 100	band	evidence level

- **Reports:** online eval; same-base SFT-only baseline; step-budget control; data-scale ablation; detector check; code announced.
- **Flags:** none recorded. **Not reported:** std or CI; held-out apps unclear.
- Pattern R03 (mine screen videos). Authors' numbers, not rerun.

Method: label videos with an inverse dynamics model, then train in two stages

- **Sees / acts:** last 5 screenshots + history in; click, type, scroll, key calls out.
- **Data:** 39,000 videos give 1.52M steps, each with a short thought.
- **Signal:** IDM-labelled steps (stage 1), then human trajectories (stage 2).



Our redraw (not the paper's figure): the labelling pipeline, two training stages and the resulting agent loop.

$$f_\theta(v) \rightarrow \{(a_k,\, t_k^s,\, t_k^e)\}_{k=1}^{K}$$

Dense event detection: typed GUI events with start and end times

Key results: OSWorld-Verified task success (%), online

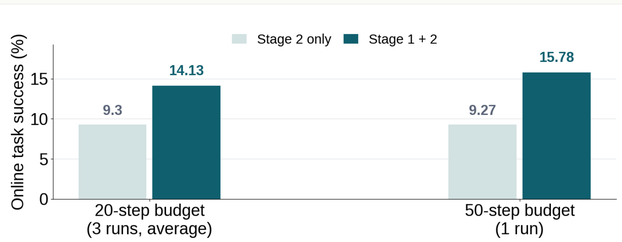


Table 4, redrawn: Qwen2.5-VL-7B; 20 steps = mean of 3 runs, 50 steps = 1 run.

Stage 2 only stays at 9.27 at 50 steps; stage 1 + 2 reaches 15.78

SO WHAT → *Offline AgentNetBench step SR (not charted): 69.3 vs 64.1 SFT-only, 38.5 base.*

Data scale: OSWorld-Verified (50 steps)

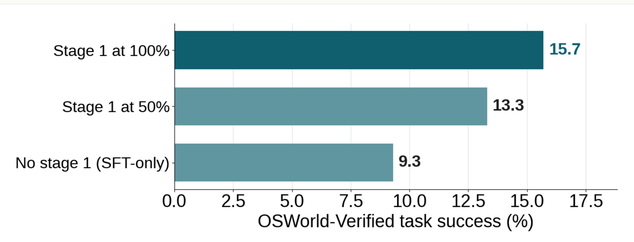


Figure 6, redrawn: share of stage-1 video tokens, same stage 2.

SO WHAT → *Success at 50 steps rises with stage-1 data: 9.3, 13.3, then 15.7. Offline AgentNetBench step success rises 64.1, 68.1, 69.3 over the same three settings. The detector is checked on 23 held-out hours (precision 0.88, recall 0.70); predicted action parameters were judged correct for 0.658 of 500 manual samples (drag 0.366, press 0.362). Stage 2 alone stays at 9.27 at 50 steps.*

Headline Numbers

15.8%
OSWorld-Verified SR
vs 9.3% SFT-only; 50 steps

- **1.52M** mined action steps
- **39,000** YouTube videos
- **69.3%** AgentNetBench, offline

The paper's own controls

CONTROL	REPORTED	COMPARED WITH
Steps 20 → 50	14.13 → 15.78	SFT-only 9.3 → 9.27
Three 20-step runs	13.57 – 14.68	SFT-only 9.13 – 9.42
AgentNetBench (offline)	69.3	SFT-only 64.1
Detector, held-out 23 h	P 0.88 / R 0.70	20,282 events

Takeaway

Labels from a learned inverse dynamics model on 39,000 tutorial videos are reported to lift OSWorld-Verified from 9.3% to 15.8% on a 7B base.

1.52M mined steps · 26B stage-1 tokens · stage 1: 32×H100, ~60 h

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Paper · Code